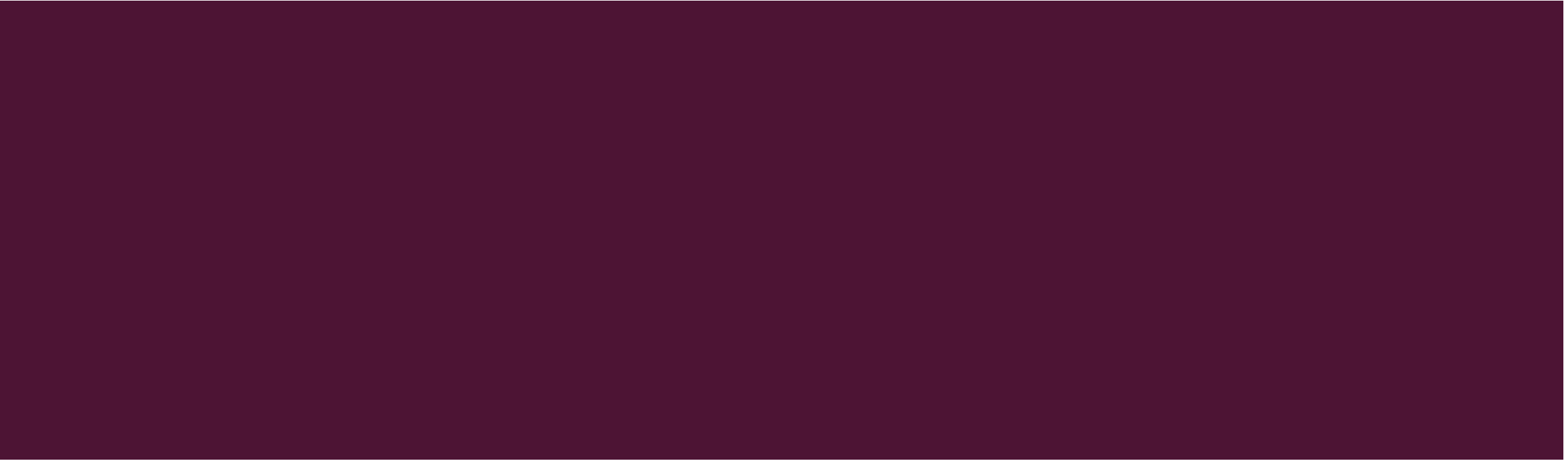

TEAM MINI PROJECT 4: WRITING SQL QUERIES

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DS-450



SINGLE TABLE QUERIES

1. Get a list of all film titles alphabetized by title.

```
SELECT title  
FROM film  
ORDER BY title;
```

	title
▶	ACADEMY DINOSAUR
	ACE GOLDFINGER
	ADAPTATION HOLES
	AFFAIR PREJUDICE
	AFRICAN EGG
	AGENT TRUMAN
	AIRPLANE SIERRA
	AIRPORT POLLOCK

SINGLE TABLE QUERIES

2. Find the description, release year, length, and rating for the movie “KENTUCKIAN GIANT”.

```
SELECT title, description, release_year, length, rating
FROM film
WHERE title = "KENTUCKIAN GIANT";
```

	title	description	release_year	length	rating
▶	KENTUCKIAN GIANT	A Stunning Yarn of a Woman And a Frisbee who...	2006	169	PG

SINGLE TABLE QUERIES

3. Find the first name and last name of each employee (staff table). Your query should include the last name first, and then the first name.

```
SELECT last_name, first_name  
FROM staff;
```

	last_name	first_name
▶	Hillyer	Mike
	Stephens	Jon

SINGLE TABLE QUERIES

4. Repeat the query above, but this time, the results should include only one column with the format last name, first name. The output column should be named “name”.

```
SELECT CONCAT(last_name, ', ', first_name) AS name  
FROM staff;
```

	name
▶	Hillyer, Mike
	Stephens, Jon

SINGLE TABLE QUERIES

5. Get the number of customers. The output should be a single number. Name the column “num_customers”.

```
SELECT COUNT(*) AS num_customers  
FROM customer;
```

	num_customers
▶	599

SINGLE TABLE QUERIES

6. Get the number of customers who are active vs inactive in the system.

```
SELECT active, COUNT(*) AS count  
FROM customer  
GROUP BY active;
```

	active	count
▶	1	584
	0	15

SINGLE TABLE QUERIES

7. Get the average amount a customer spends on a rental.

```
SELECT AVG(amount) AS avg_spent  
FROM payment;
```

	avg_spent
▶	4.201356

SINGLE TABLE QUERIES

8. Get maximum amount any customer has spent on a rental.

```
SELECT MAX(amount) AS max_spent  
FROM payment;
```

	max_spent
▶	11.99

SINGLE TABLE QUERIES

9. Get a list of the actors. The results should include only one column with the format last name, first name. The column should be named “actor_name” The results should be sorted alphabetically by the last name (ascending).

```
SELECT CONCAT(last_name, ', ', first_name) AS actor_name
FROM actor
ORDER BY last_name;
```

	actor_name
▶	AKROYD, KIRSTEN
	AKROYD, CHRISTIAN
	AKROYD, DEBBIE
	ALLEN, MERYL
	ALLEN, KIM
	ALLEN, CUBA
	ASTAIRE, ANGELINA
	BACALL, RUSSELL

SINGLE TABLE QUERIES

10. Repeat this query above, but the results should be in reverse order.

```
SELECT CONCAT(last_name, ', ', first_name) AS actor_name  
FROM actor  
ORDER BY last_name DESC;
```

	actor_name
▶	ZELLWEGER, CAMERON
	ZELLWEGER, JULIA
	ZELLWEGER, MINNIE
	WRAY, CAMERON
	WOOD, UMA
	WOOD, FAY
	WITHERSPOON, ANGELA
	WINSLET, RIP

MULTI-TABLE QUERIES

1. Get a list of category names and a count of movies that fall into that category. Name the category column “category” the count column “num_films”. Order the results alphabetically (ascending). Use the WHERE clause to join the tables.

```
select name as category, Count(*) as num_films
from category, film, film_category
where film_category.film_id = film.film_id and film_category.category_id = category.category_id
group by name
order by name;
```

2. Repeat the query above using a JOIN clause instead of the WHERE clause.

```
select name as category, Count(*) as num_films
from film_category join category on film_category.category_id = category.category_id
join film on film_category.film_id = film.film_id
group by name
order by name;
```

category	num_films
Action	64
Animation	66
Children	60
Classics	57
Comedy	58
Documentary	68
Drama	62

MULTI-TABLE QUERIES

3. Get a list of country names and a count of the cities that are in that country. Name the count column “num_cities”. Order the results alphabetically (ascending). Use the WHERE clause to join the tables.

```
select country as country, Count(*) as num_cities
from country, city
where city.country_id = country.country_id
group by country
order by country asc;
```

4. Repeat the query above using a JOIN clause instead of the WHERE clause.

```
select country as country, Count(*) as num_cities
from country join city on city.country_id = country.country_id
group by country
order by country;
```

country	num_cities
Afghanistan	1
Algeria	3
American Samoa	1
Angola	2
Anguilla	1
Argentina	13
Armenia	1
Australia	1
Austria	3
Azerbaijan	2
Bahrain	1
Bangladesh	3

MULTI-TABLE QUERIES

5. Get a list of each customer's last name and first name and the number of rentals they have. Name the count column "num_rentals". Order the result by the number of rentals in descending order. The highest number of rentals should be at the top. Sort any es (same number of rentals) by last name (ascending). Use the WHERE clause to join the tables.

```
select last_name, first_name, Count(*) as num_rentals
from customer, rental
where rental.customer_id = customer.customer_id
group by last_name, first_name
order by num_rentals desc, last_name asc;
```

6. Repeat the query above using a JOIN clause instead of the WHERE clause.

```
select last_name, first_name, Count(*) as num_rentals
from customer join rental on rental.customer_id = customer.customer_id
group by last_name, first_name
order by num_rentals desc, last_name asc;
```

last_name	first_name	num_rentals
HUNT	ELEANOR	46
SEAL	KARL	45
DEAN	MARCIA	42
SHAW	CLARA	42
SANDERS	TAMMY	41
BULL	WESLEY	40
PETERS	SUE	40
CARY	TIM	39
KENNEDY	RHONDA	39
SNYDER	MARION	39
BATES	DAISY	38
BROWN	ELIZABETH	38

MULTI-TABLE QUERIES

7. Get a list of each customer's last name and first name and the amount of money they have spent on rentals. Name the sum column "total_spent". Order the result by the amount in descending order. The highest amount of money spent should be at the top. Sort any ties (amount of money spent) by last name (ascending). Use the JOIN clause for this query.

```
select last_name, first_name, sum(payment.amount) as total_spent
from customer join payment on payment.customer_id = customer.customer_id
group by last_name, first_name
order by total_spent desc, last_name asc;
```

last_name	first_name	total_spent
SEAL	KARL	221.55
HUNT	ELEANOR	216.54
SHAW	CLARA	195.58
KENNEDY	RHONDA	194.61
SNYDER	MARION	194.61
COLLAZO	TOMMY	186.62
BULL	WESLEY	177.60
CARY	TIM	175.61
DEAN	MARCIA	175.58
BRADLEY	ANA	174.66
CARROLL	JUNE	173.63
COLLINS	DIANE	169.65

MULTI-TABLE QUERIES

8. Get the number of actors in each film. Order the results (ascending) by the film title and name the column with the actor count “num_actors”.

```
select title as film, count(actor_id) as num_actors
from film join film_actor on film_actor.film_id = film.film_id
group by title
order by title;
```

film	num_actors
ACADEMY DINOSAUR	10
ACE GOLDFINGER	4
ADAPTATION HOLES	5
AFFAIR PREJUDICE	5
AFRICAN EGG	5
AGENT TRUMAN	7
AIRPLANE SIERRA	5
AIRPORT POLLOCK	4
ALABAMA DEVIL	9
ALADDIN CALENDAR	8
ALAMO VIDEOTAPE	4
ALASKA PHANTOM	7

MULTI-TABLE QUERIES

9. Get the number of films each manager holds. Use only the manager staff id to identify the manager. Name the column with the number of films “num_films”.

```
select manager_staff_id as manager, count(film_id) as num_films
from store join inventory on inventory.store_id = store.store_id
group by manager_staff_id
order by manager_staff_id;
```

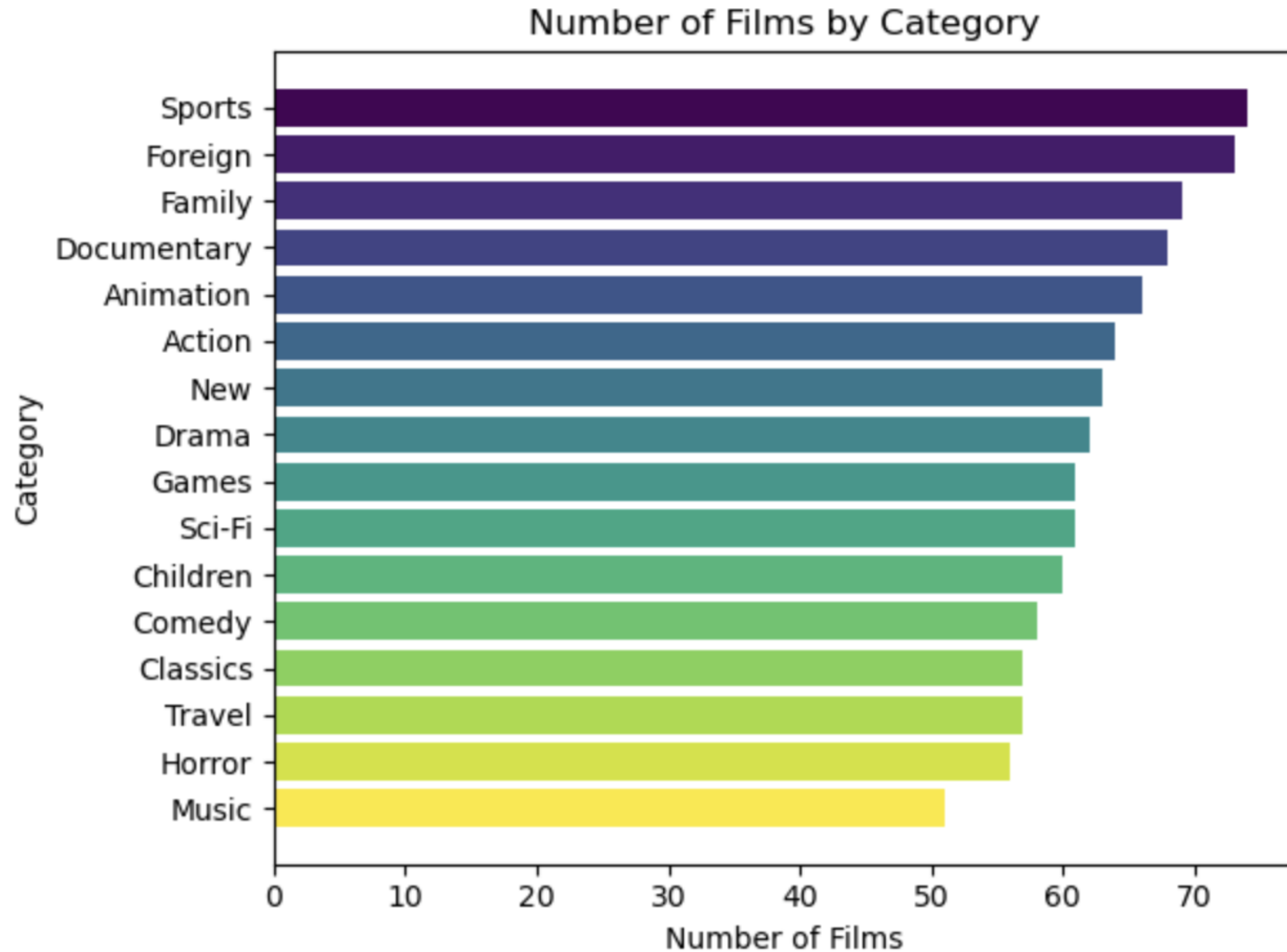
manager	num_films
1	2270
2	2311

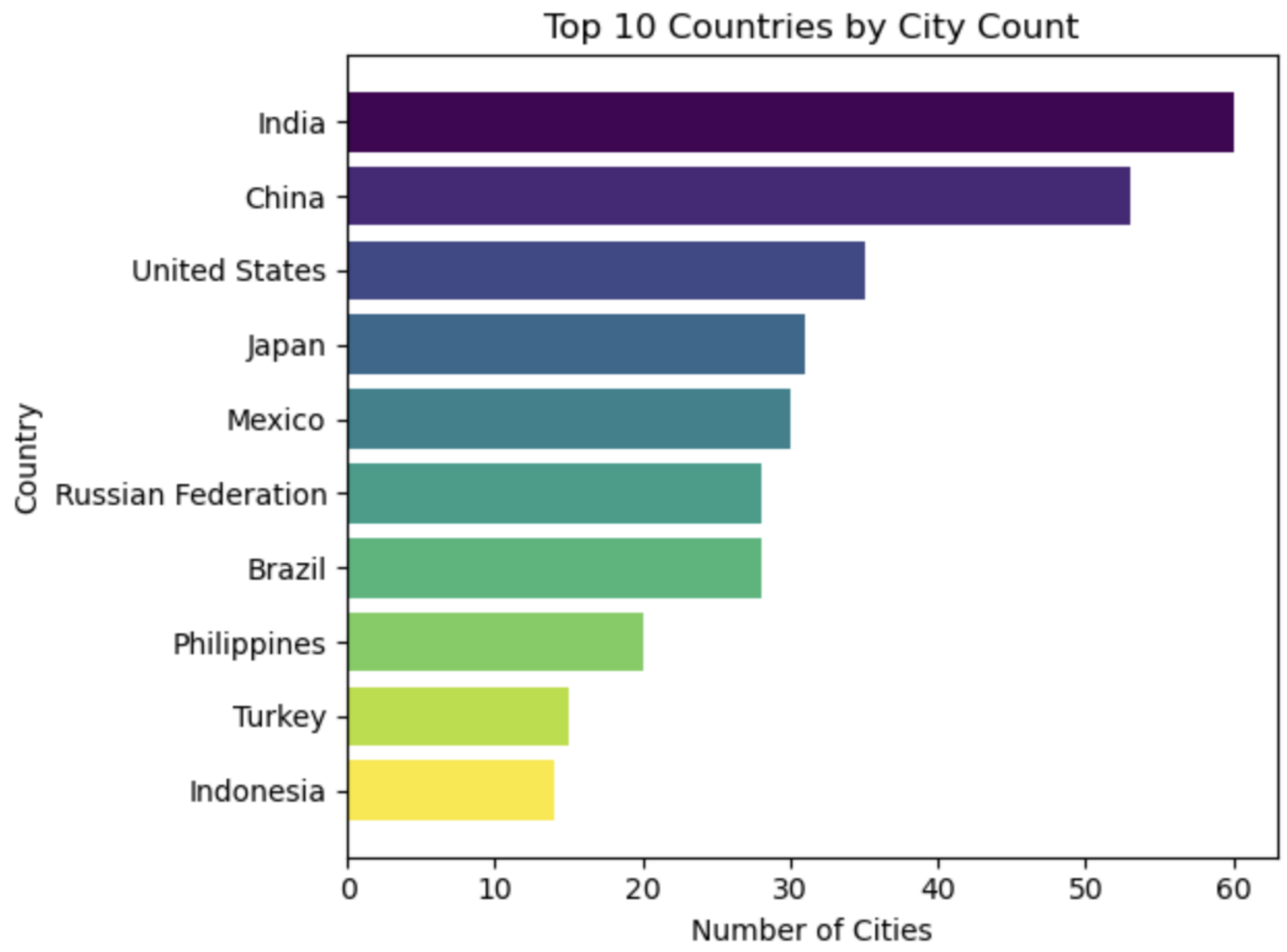
MULTI-TABLE QUERIES

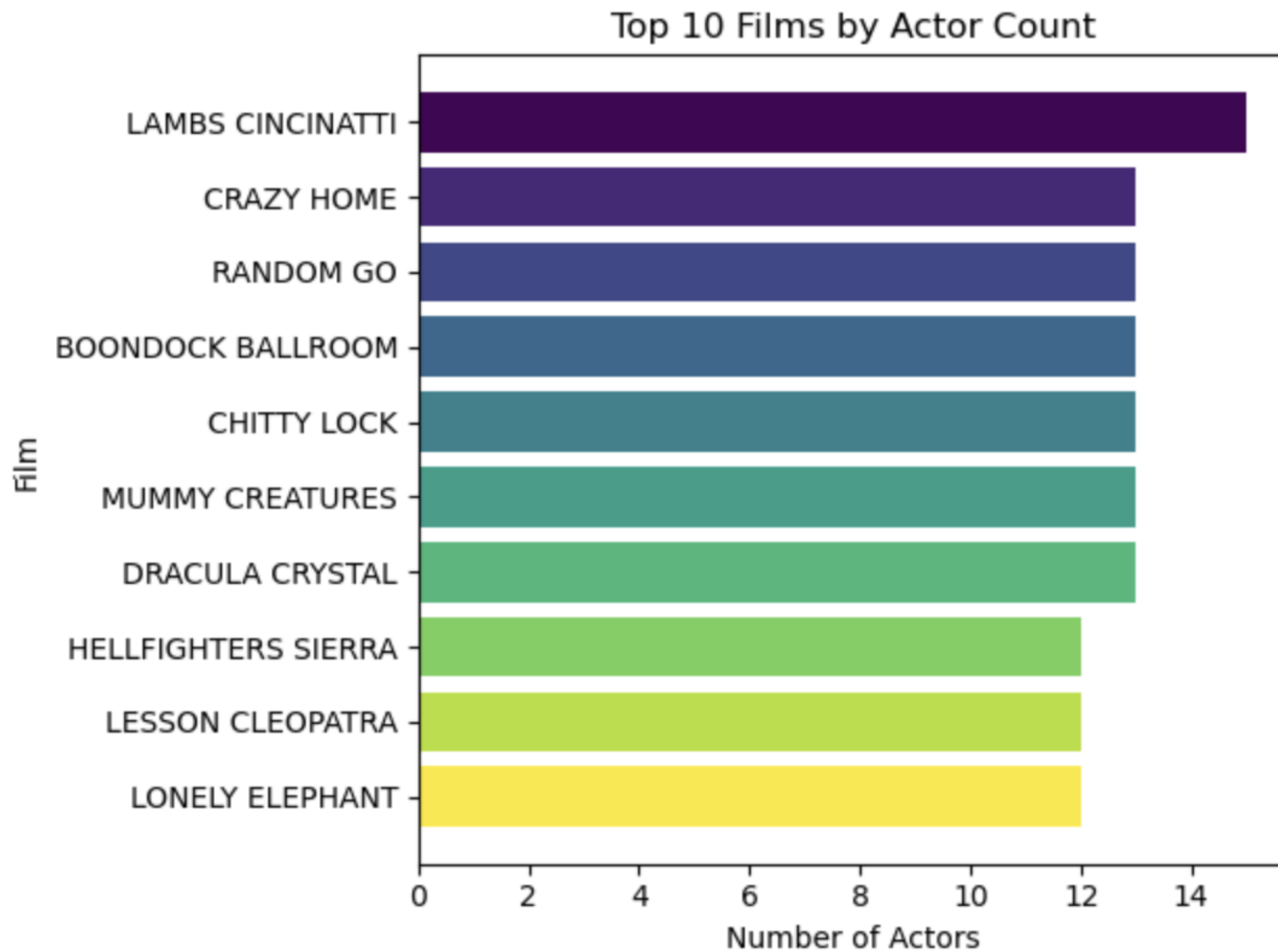
10. Get the number of customers per manager. Use only the manager staff id to identify the manager. Name the column with the number of films “num_customers”. Order by store id (ascending).

```
select manager_staff_id as manager, count(customer_id) as num_customers
from store join customer on customer.store_id = store.store_id
group by manager_staff_id
order by store.store_id asc;
```

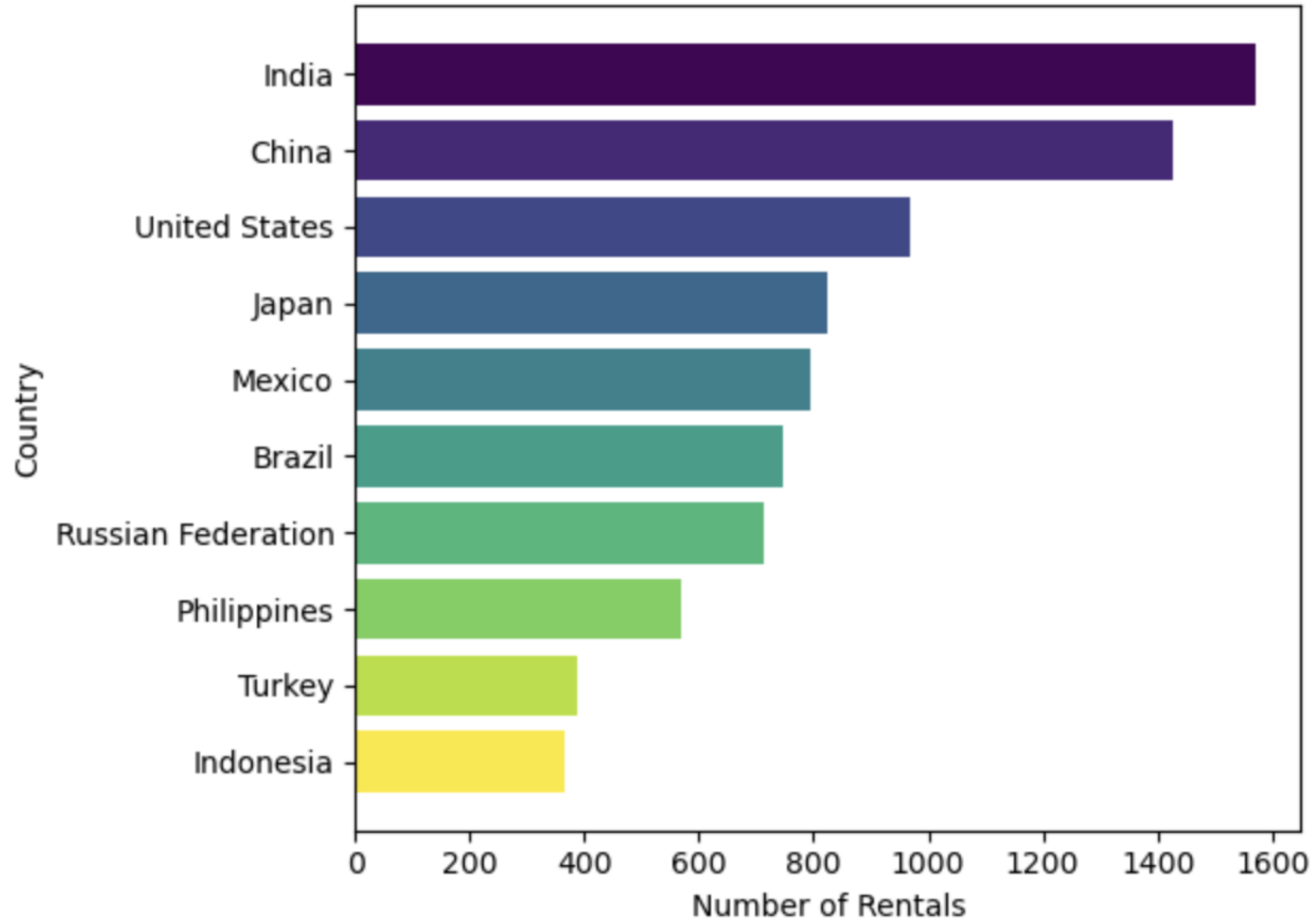
manager	num_customers
1	326
2	273





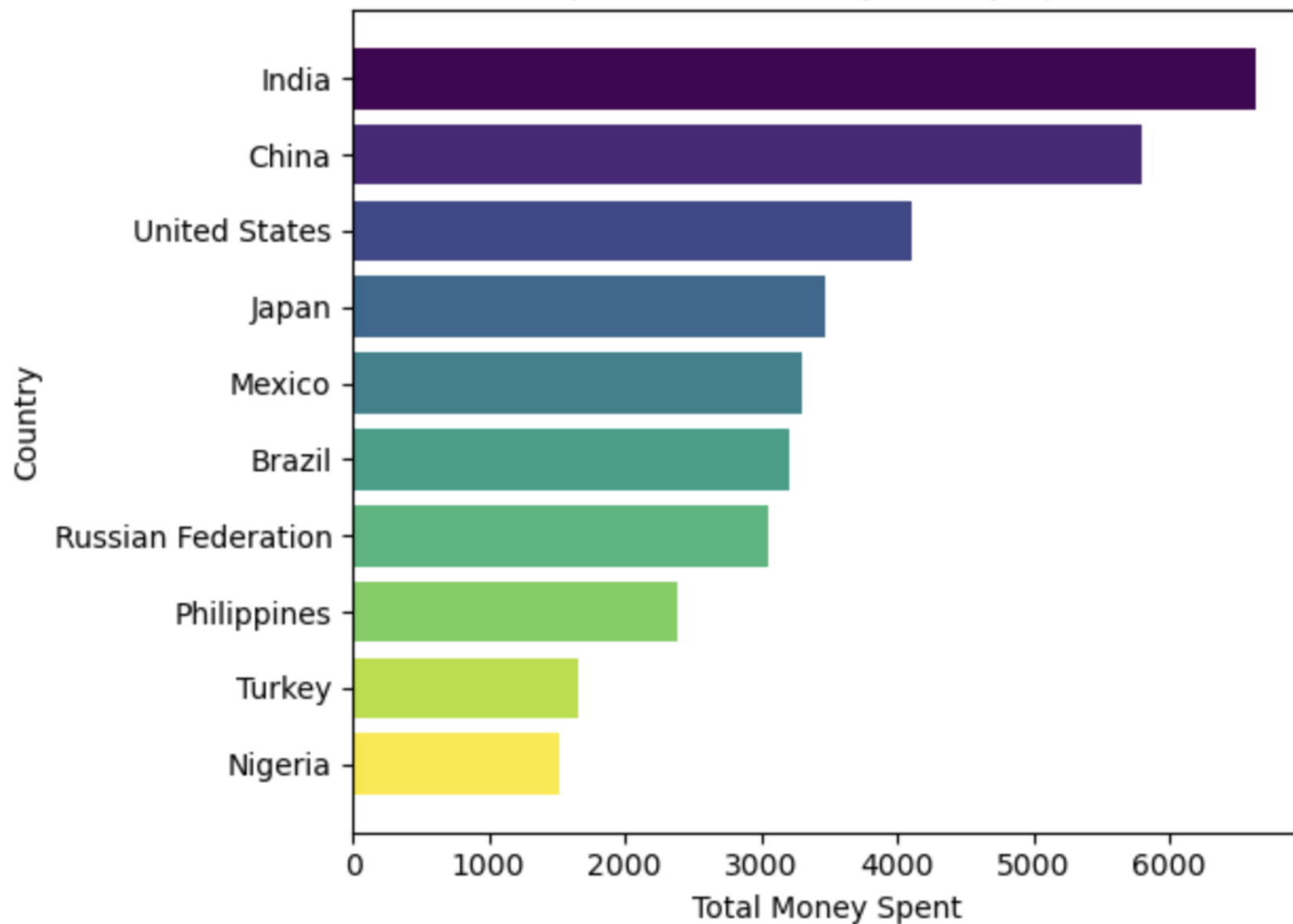


Top 10 Countries by Rentals

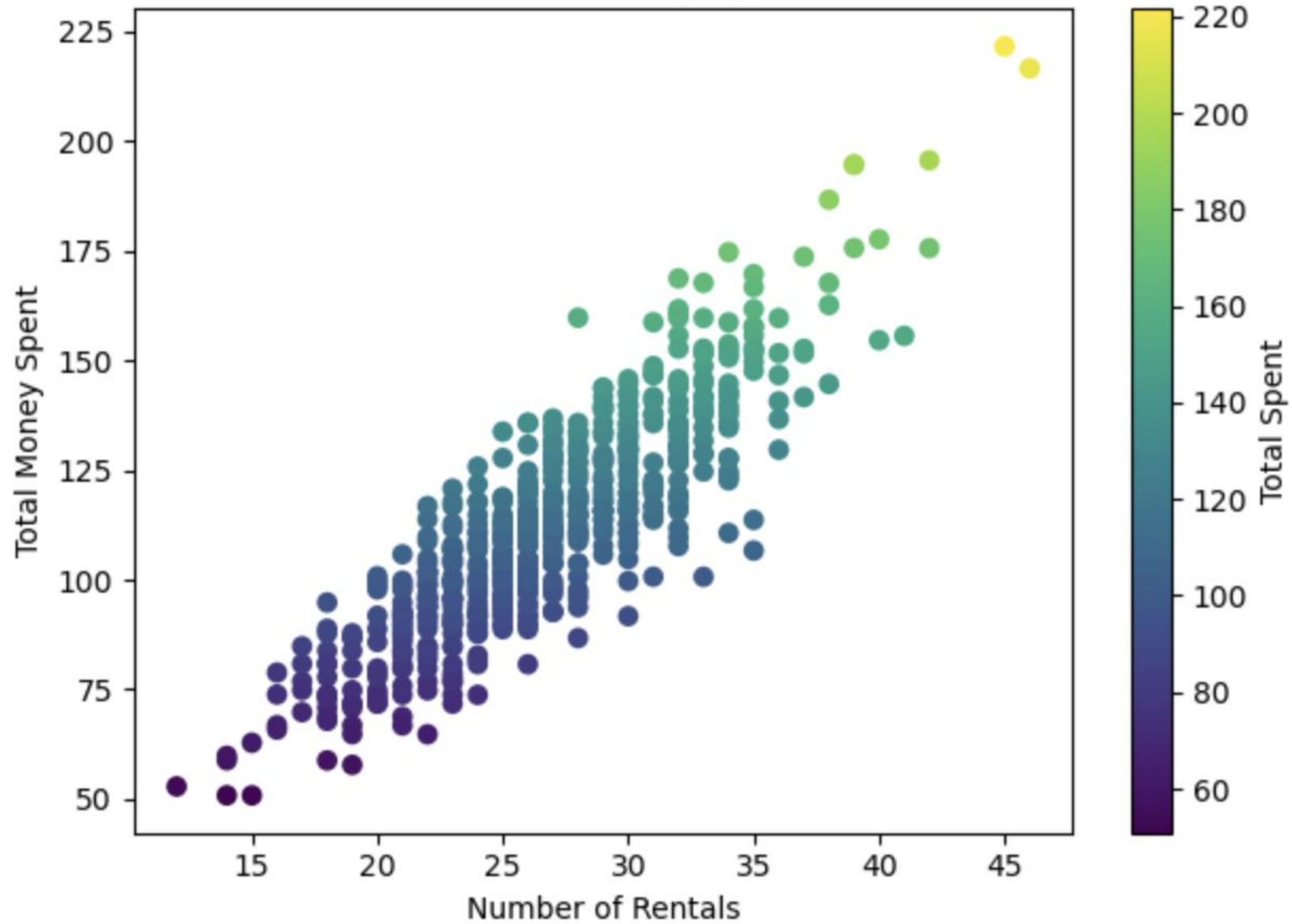




Top 10 Countries by Money Spent



Rentals vs Money Spent



RELATIONSHIP BETWEEN # OF RENTALS & MONEY SPENT

Slope: 4.3216392141972495

Intercept: -3.2217354801012874

R value: 0.8706639553973844

R-squared: 0.7580557232282186

P-value: 4.099419064718929e-186